

Summary: A Taxonomy of Generative AI Use for Course-Related Independent Learning

Olmanson, J., Hassani, A., & Jeon, M. (2025)

Core Framework: The Six-Category Taxonomy

The study presents an empirically-derived taxonomy mapping how undergraduate and graduate students use generative AI in course-related independent learning (CRIL)—defined as "learner-managed work with which students engage outside of scheduled class meeting times to build and demonstrate knowledge aligned with course curricular goals" (p. 3). The taxonomy progresses along a continuum of learner agency:

Not for Me (avoidance due to ethics, cost, or lack of knowledge), **Escape** (using AI to complete tasks with minimal effort), **Get Me Going** (building momentum through planning or unsticking), **Feedback Please** (seeking external review), **Help Me Learn** (tutoring and conceptual support), and **Magnify** (amplifying creativity and ambition beyond assignment expectations).

"Together they map a continuum of learner agency ranging from avoidance to amplification" (p. 12).

The authors note that "undergraduate participants tended to describe generative AI CRIL usage in a product-centric, externalized way (e.g., check my work, do it all for me, teach me), whereas graduate participants more frequently positioned AI as part of a process that was more formative and internal (e.g., give me feedback, help me learn)" (p. 11).

The Escape Paradox

The most problematic yet nuanced category distinguishes between Escape Level 1 (avoiding core learning due to poor time management, low motivation, or feeling overwhelmed) and Escape Level 2 (delegating "tedious" or "peripheral" tasks students judge won't contribute to learning). The researchers found graduate students more likely to articulate strategic rationales for Level 2 use, with one participant noting: "It's really useful when it's just tedious programming tasks that I know frontwards and backwards" (p. 15).

The study provocatively observes that "getting a generative AI application to do their work for them with their voice and in a way that minimized detection often required the highest level of iterative interaction with AI across the taxonomy—suggesting high levels of AI literacy relative to the task and metacognitive awareness adjacent to the nature of the assignment" (p. 23). This complicates simplistic narratives about "cheating" by revealing sophisticated strategic thinking even in ethically questionable uses.

Productive Struggle and AI Scaffolding

The authors frame learning as necessarily involving "productive struggle"—building on Vygotsky, Piaget, and Dewey to argue that "struggle becomes productive when students sustain their efforts to make sense of new or challenging concepts and material" (p. 3). They wrestle with AI's dual potential: it can scaffold learning within zones of proximal development, but also offers "students avenues of assignment completion that circumvent effort and reflection" (p. 5).

"Several participants noted that these same usage patterns that enabled more efficient and satisfying progress also acted as a potential gateway toward expanded Escape-like use—as the LLMs, like an over-eager intern, often anticipated next steps and offered to carry them out on behalf of the student" (p. 24).

Methodological Contribution: Learner-Initiated AI Use

Unlike prior AI-in-education research focused on systems that "predict learning outcomes, intervene of its own accord on the learner, or both," this study examines learner-initiated interactions (p. 2). The authors synthesized qualitative data from 25 students (7 graduate, 18 undergraduate) across two design-based research studies, using abductive reasoning to iteratively develop and refine the taxonomy through five phases including stakeholder feedback from K-12 and higher ed contexts.

"Rather than viewing generative AI as an instructional intervention that acts on learners, the taxonomy positions students as initiators, leaders, and orchestrators of their own CRIL experiences and processes" (p. 25).

Implications for Practice

The taxonomy offers diagnostic potential for curriculum design. Educators should inventory assignments to identify which might fall under "Escape" (busy work, perceived as tedious) versus which invite Magnify-level engagement. The authors call for institutional policies that move beyond "integrity risk" framing toward treating "AI as a potential collaborator and define originality as transparent co-creation rather than solitary production" (p. 28).

For students, the taxonomy functions as a reflective tool: "By mapping their own AI use along the continuum, students may notice that they tend to use, or conceive of AI use in CRIL, as only involving Escape. This realization may lead to their strategic experimentation with uses in other categories" (p. 27). The authors suggest students track their affective states—using AI when feeling "motivated, creative, confident" versus "desperate, reckless, or indifferent" (p. 27).

Relevance to Your Research: Critical Pedagogy Meets AI Taxonomy

This taxonomy offers significant analytical leverage for your dissertation work at the intersection of critical pedagogy, educational technology, and power structures in learning. Several dimensions align powerfully with your Freirean framework:

Power and Agency Dynamics

The taxonomy's vertical axis—literally labeled "agency" in Figure 3—makes visible how different AI uses redistribute power in the learning relationship. "Escape" represents students reclaiming time/energy from what they perceive as oppressive "busy work," while "Magnify" shows learners transcending institutional constraints on their intellectual ambition. You could analyze this through Freire's banking model critique: does CRIL itself reproduce teacher-as-depositor dynamics that students resist via AI? The finding that graduate students frame AI use more "internally" and "formatively" versus undergraduates' "externalized" framing (p. 11) suggests developmental shifts in critical consciousness about learning.

Decolonizing Assessment and Originality

The authors' call to redefine originality as "transparent co-creation rather than solitary production" (p. 28) directly challenges Western individualist epistemologies embedded in academic integrity policies. This connects to your work on indigenous knowledge systems—the Arakbut study you analyzed emphasized collective, relational knowing. You could argue that current AI policies enforce colonial notions of autonomous authorship that contradict both indigenous pedagogies AND the actual socially-distributed nature of cognition. The taxonomy provides empirical evidence that students are already operating in collaborative human-AI knowledge production; institutions need decolonial frameworks to legitimize this rather than police it.

Methodological Resonance

Their five-phase abductive approach—integrating participant observations, iterative coding, and stakeholder validation—models the kind of participatory, dialogical research Freire advocated. For your autoethnography project and tech-based website, you could adapt their taxonomy as a reflective prompt structure, asking educators to map their own AI use across categories and critically examine what each reveals about their pedagogical stance and relationship to institutional power. Their distinction between "learner-initiated" AI use versus system-imposed intervention (p. 2) parallels your emphasis on student agency and resistance to top-down educational design.

Theoretical Gaps to Critique

While rich empirically, the taxonomy remains under-theorized from critical perspectives. They cite Vygotsky on ZPD but don't engage with critiques of how scaffolding can reproduce dependency. They note "busy work" perceptions but don't examine whose interests course designs serve—a key Freirean question. The "productive struggle" frame (p. 3) risks valorizing difficulty without interrogating

whether struggle serves learning or institutional gatekeeping. Your dissertation could extend this taxonomy by overlaying critical pedagogy questions: Struggle for whom? Agency toward what ends? Which uses reproduce versus resist neoliberal educational logics?

Practical Application

The taxonomy's diagnostic questions for educators (pp. 27-28) could be reframed through your critical lens. Instead of "How might these assignments be adjusted to decrease Escape use?" ask "What does Escape reveal about whose knowledge and labor we value?" The authors suggest students track affective states (p. 27)—you could develop this into a critical consciousness-raising tool where students analyze why certain tasks trigger desperation versus motivation, connecting individual experience to structural educational inequities. This would operationalize your Freirean approach of moving from naive to critical consciousness through dialogue and problem-posing about technology.